

4 (short)

A closed ended cylindrical pressure vessel is shown in Fig. 4. The circumferential walls of the pressure vessel are fabricated from 4 mm thick steel and have an outside diameter of 200 mm. The pressure vessel supports an external vertical load of 123 kN applied concentrically through the closed end plates. If the internal pressure is such that the stress in the longitudinal direction is zero, what is the stress in the circumferential direction? Clearly state any assumptions made. [10]



Fig. 4